

Kadircan Kara, M. Sc. – Electrical & Electronics Engineer

📍 Istanbul, Turkey

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PROFILE

Motivated M.Sc. graduate in Computer Science with a strong foundation in multi-objective optimization. Skilled in Python and MATLAB. Passionate about implementing operations research methods in real-world problems.

EDUCATION

Master of Science in Computer Science Spring 2022 – Fall 2025
Ozyegin University, Istanbul, Turkey
Thesis: Multi-UAV Cooperative Search Under Sensing and Communication Limitations
GPA: 3.67/4.00

Bachelor of Science in Electrical and Electronics Engineering Fall 2016 – Spring 2021
Ozyegin University, Istanbul, Turkey

Erasmus Exchange Fall 2019
Universita Degli Studi di Napoli Federico II, Napoli, ITALY

WORK EXPERIENCE

Customer Success / Technical Support Specialist - *Part-time* Oct 2020 – Nov 2023

+ ***Company: Faradai, Istanbul, Turkey***

- ◆ Provided management for customers with periodic portfolio reports, meetings and weekly mailings which resulted in more than **10% financial savings**.
- ◆ Designed alarm scenarios for real-time equipment monitoring and abnormal activity detection using JavaScript-based console.
- ◆ Finalized side-projects in *Python* that improved internal processes, **saving more than 15 person-hours per week**.
- ◆ Prepared documentations on the alarm system in both English and Turkish.
- ◆ Trained, coached and supervised new staff members.

Researcher - *Internship*

Aug – Sep 2019

+ *Company: EPIAŞ, Istanbul, Turkey*

- ◆ Processed and analyzed large-scale hourly and seasonal energy consumption data using MATLAB.
- ◆ Forecasted future energy consumption data using classification and clustering algorithms.

PROJECTS

MSc. Thesis Research

Spring 2022 – Fall 2025

+ *Stack: Python (PyMOO, Numpy, Matplotlib)*

- ◆ Designed and evaluated multi-UAV cooperative search algorithms. Work led to 2 IEEE conference publications (see Publications).

Turtlebot 3 Autonomous Navigation with Object Avoidance

Sep 2020 – Feb 2021

+ *Stack: Linux, Raspberry Pi, ROS, SLAM, Gazebo, Rviz*

- ◆ Designed and implemented a ROS-based autonomous robot capable of locating a target and navigating to an exit using SLAM without image recognition.
- ◆ Built and tested the full robotic workflow in Gazebo and visualized mapping, localization, and navigation performance using RViz.
- ◆ Deployed the system on Raspberry Pi running Linux, optimizing real-time performance and mobility for the entire operation duration.
- ◆ Achieved mission objectives of fast target reach, continuous mobility, and zero collision, demonstrating end-to-end autonomous behavior.

BMS Optimization

Sep 2019 – Feb 2020

+ *Stack: Simetrix, Vivado HLS*

- ◆ Improved the BMS in electric vehicles using Lithium – Ion batteries utilizing Simetrix for the optimization of the circuitry and Vivado HLS for implementing the system on chip (SoC).

Day-Ahead Energy Market (DAM) Clearing Problem

Feb – June 2019

+ *Stack: MATLAB, Cplex*

- ◆ Developed an optimization algorithm for the Day-Ahead Electricity Market (DAM) to maximize energy market surplus (social welfare).
- ◆ Designed and implemented a MATLAB-based GUI to visualize bids, market parameters, and optimization outputs for improved user interaction.
- ◆ Built a market-clearing model that supports hourly, block, and flexible bids, handling diverse bidder constraints for realistic DAM conditions.

OzU Microgrid with Renewable Energy Generation and DSM Techniques Feb – June 2020

+ **Stack: MATLAB**

- ◆ Designed a microgrid for Ozyegin University which generates its own electricity.
- ◆ Analyzed recent annual energy consumption data to decide the quantities for solar panels and wind turbines.
- ◆ Implemented demand side management techniques to lower energy consumption.
- ◆ Presented the feasibility analysis and details about ROI of the project.

A.S.C.I.D.

Feb – June 2019

+ **Stack: Arduino**

- ◆ Built a sound sensor system using Arduino with the help of a digital microphone and 3 LEDs
- ◆ The microphone's detected decibel level determines which LED lights up: green for low volume, yellow for medium, and red for high.

Generator Operating Cost Minimizer

Sep 2018 - Feb 2019

+ **Stack: MATLAB**

- ◆ Implemented the lambda-iteration algorithm to minimize generator operating costs with the help of a graphical user interface (GUI) designed with MATLAB.

Automated Tendering Process

June – August 2021

+ **Stack: Python (Dash, Pandas, Numpy libraries)**

- ◆ Designed a small-scale web application using Dash that generates a tendering document given input parameters by the user, considering the current currency exchange rates, and price list.

SKILLS

Soft Skills: Data analysis, Mathematical modelling, Data visualization, Algorithm design

Programming: Python (OOP), MATLAB, SQL, C++, Verilog, Java, JavaScript, Vivado HLS

Version Control: Git

OS: Linux

Circuit Design: LTSpice, Simetrix

RESEARCH & PUBLICATIONS

Cooperative Multi-Target Search with UAV

Swarms: Evolutionary vs. Reinforcement Learning Strategies

September 2025

+ *Venue: IEEE MSWiM 2025*

- ◆ Conducted a comparative analysis of evolutionary optimization and reinforcement learning in cooperative UAV missions, emphasizing detection and communication efficiency.

Joint Optimization of Coverage, Connectivity and Revisit Time in Multi-UAV Path Planning

June 2025

+ *Venue: IEEE VTC 2025*

- ◆ Proposed a novel joint multi-objective optimization framework to generate multi-UAV paths under constraints.

LANGUAGES

Turkish: Native

English: Advanced (C1) – TOEFL 88 (Dec 2021)